

JAY AGARWAL

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Delhi - 110041, India

EDUCATION

• Birla Institute of Technology and Science, Pilani

November 2022 - August 2027

M.Sc. Physics and B.E. Computer Science

Pilani, India

- CGPA: 8.1/10.00
- Relevant Coursework: Classical Mechanics, Electromagnetic Theory I & II, Quantum Mechanics I & II, Quantum Info and Computation, Computational Physics, Atomic & Molecular Physics, Nuclear & Particle Physics, Group Theory, Theory of Relativity, Data Structures & Algorithms, Deep Learning, Introduction to LLMs, Robotics

RESEARCH EXPERIENCE

• Undergraduate Research – Symmetry-Aware Models for HEP

October 2025 - Present

Advisor: Prof. Dhruv Kumar | Tools: Python, PyTorch, PyG, Uproot

BITS Pilani

- Evaluated modern machine learning techniques for High Energy Physics (HEP) tasks, including Jet Tagging, Amplitude Regression, and Event Generation.
- Conducted an in-depth architectural analysis of equivariant models, specifically focusing on the Lorentz Equivariant Geometric Algebra Transformer (LGATr).
- Applied mechanistic interpretability techniques to LGATr, extracting internal representations via attention score distributions, heatmaps and linear probing; released as [arXiv:2606.21790](https://arxiv.org/abs/2606.21790).
- Implementing knowledge distillation to compress the equivariant LGATr into a standard non-equivariant baseline, systematically evaluating performance and accuracy trade-offs.

• Research Collaborator – TAMBO Neutrino Observatory

April 2026 - Present

Advisor: Prof. Tommaso Dorigo | Tools: Python, PyTorch, Differentiable Programming

INFN Padova (Remote)

- Optimizing detector layout for the TAMBO neutrino observatory, formulating geometry and instrumentation choices as differentiable objectives and applying gradient-based end-to-end optimization over the design space.
- Performed landscape analysis over the design space and benchmarking surrogate models against full-simulation ground truth.

• Undergraduate Research - LARR Dynamics in Intense Laser Fields

August 2025 - December 2025

Advisor: Prof. Amol R. Holkundkar | Tools: Python, SciPy, Flint, Multiprocessing

BITS Pilani

- Simulated Laser-Assisted Radiative Recombination (LARR) dynamics for hydrogenic ions driven by intense few-cycle laser pulses, evaluating full quantum-mechanical and semiclassical emission probabilities.
- Optimized computation of high-dimensional complex integrals and confluent hypergeometric functions using Python multiprocessing and shared-memory arrays.
- Conducted systematic Carrier-Envelope Phase (CEP) sweeps; generated heatmaps replicating literature spectral cutoff features and revealing CEP-dependent asymmetries.

• Solid State Physics Lab, DRDO

May 2024 - July 2024

Research Intern

Delhi, India

- Worked in the Quantum Sensing Group on atomic frequency standards, focusing on atom-light interactions and Coherent Population Trapping (CPT) in alkali atoms.
- Simulated CPT signal behavior and assisted experimental efforts to optimize cavity wall surface roughness in atomic clock setups.
- Developed a Python-based GUI tool to model absorption and transmittance spectra with adjustable parameters; achieved high-accuracy CPT predictions aligned with experimental data and the tool was integrated into the group's ongoing research workflow.

INDUSTRY EXPERIENCE

• Wells Fargo

May 2026 - July 2026

Quantitative Analyst Intern, Causal Inference Team

Bengaluru, India

- Automated the marketing model development pipeline for the Causal Inference team, replacing repetitive manual setup, review, debugging and diagnostics with a standardized reproducible workflow.
- Designed the workflow around a common starting point for all analyses while preserving room for custom extensions, cutting turnaround time and shifting analyst effort toward modelling decisions.

PROJECTS

• Hallucination Detection in LLMs via Causal Dynamic Routing

January 2026 - Present

Tools: Python, PyTorch, LLaMA | Advisor: Prof. Dhruv Kumar

- Proposed an inference-time framework combining Semantic Entropy Probes on hidden states with Lookback Ratio attention grounding signals to detect and mitigate hallucinations without external retrieval or multiple decoding passes.
- Introduced head-adaptive gating that selectively suppresses ungrounded attention heads conditioned on predicted hallucination risk; evaluated on XSum, SQuAD, TriviaQA, and Natural Questions using LLaMA-2-7B.

• Smart Product Pricing – Amazon ML Challenge 2025

October 2025

Tools: Python, PyTorch, NumPy, Pandas, Scikit-learn

- Developed a multimodal (text + image) framework to predict product prices; designed a Gated Attention Fusion mechanism combining DeBERTa-v3 (text) and a LoRA-tuned CLIP model (image).
- Applied log transformation and standardization to normalize a long-tail price distribution, improving model stability and convergence.

• Monte Carlo Methods in Physics

October - November 2024

Tools: Python, NumPy, Matplotlib, Manim

- Implemented Metropolis-Hastings sampling to estimate the lab-frame neutrino energy spectrum from a Lorentz-boosted muon decay; validated results against analytic predictions.
- Applied Box-Muller transform and Cholesky decomposition for correlated bivariate sampling; produced an educational animation using Manim.

SKILLS

- **Programming:** Python, C, C++
- **ML & Scientific Computing:** PyTorch, JAX, Scikit-learn, NumPy, SciPy, Pandas, Matplotlib
- **Other:** Git, Linux, Manim

ACHIEVEMENTS

• Amazon ML Challenge 2025

October 2025

Amazon

- Ranked 372nd overall (peak rank 8) out of 20,000+ participating teams.

• Quantum Fall Fest

October 2025

IISc

- Ranked 5th on the overall daily challenge leaderboard.

• CodeEd Hackathon

September 2025

upGrad

- Finalist. Designed 'AdaptAI', a full-stack Next.js application converting PDFs, DOCX, video, and audio into structured educational content using AI models.

• APOGEE Game Jam '24

April 2024

BITS Pilani

- Won Special Mention among 180 teams for "Cosmic Crusade," a mixed-reality mobile game.

LEADERSHIP EXPERIENCE

- **Coordinator - Peer Mentorship Program**

June 2025 - Present

BITS Pilani

- Lead a 150-member volunteer team supporting 1,200+ incoming students. Coordinate with the Chief Warden's office and have initiated a joint mental health initiative with MPower, the campus therapy service.

- **President - Kalipatnapu AR-VR Lab**

September 2024 - September 2025

BITS Pilani

- Culmination of three years with the lab (2023–25), growing from member to President. Scaled the lab from 25 to 60 members and supervised R&D across 10+ projects.

- **Projects Chair - ICXR**

September 2025 - December 2025

Intercollegiate

- Organised a paper reading group on XR/AI research and coordinated projects within an international student community.

EXTRA CURRICULAR ACTIVITIES

- **Member - BITS Pilani Hockey Team**

BITS Pilani

- Won Gold at InterBITS and participated in the BOSM (BITS Open Sports Meet) tournament.

- **Member - The Debating Society**

BITS Pilani

- Participated in national-level tournaments, led social media and outreach for QED '23 and BPPD '24, driving record engagement.